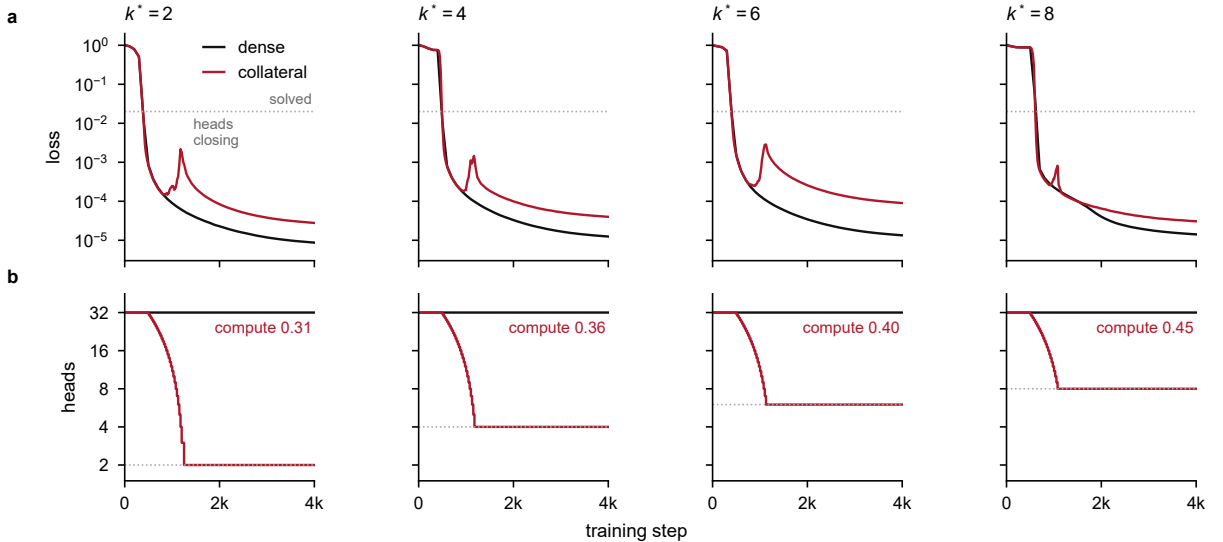




c Every method, same 30 runs

Method	Head score	During training	Gradual fade	Reopens	Gate learned	exact count never broke
Collateral rule (ours)	$\min_w L_{-h} - \min_w L$ (OBS)	yes	yes	yes	no	29/30 30/30
<i>Published methods</i>						
ZipLM-style (Kurtic et al. 2023)	$\min_w L_{-h} - \min_w L$ (OBS)	no	no	no	no	24/30 0/30
Michel et al. 2019	$ \partial L / \partial g_h $	yes	no	no	no	28/30 0/30
Learned gates (Voita et al. 2019)	$L + \lambda \sum P(g_h \neq 0)$	yes	no	no*	yes	9/30 0/30
<i>Ablations of our rule (one part removed)</i>						
no fade (instant closing)	$\min_w L_{-h} - \min_w L$	yes	no	yes	no	30/30 0/30
no re-fit (OBD score, LeCun et al. 1990)	$L_{-h}(W) - L(W)$	yes	yes	yes	no	2/30 30/30
no score (random choice)	random head	yes	yes	yes	no	30/30 6/30

Every method on the same 30 runs (tasks needing 2, 4 and 8 heads, 10 seeds each). Exact count: heads left = heads needed. Never broke: once solved, the loss never went back above the solved bar. OBS = Optimal Brain Surgeon (Hassibi & Stork 1993): the loss after removing head h , the others re-fitted. Learned gates: best of 3 penalties; *a closed gate could reopen by gradient, but none of the 2,709 closed gates did.

